

FieldNet A LoRa-Powered IoT Framework for Agricultural Automation

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Abstract—The Internet of Things (IoT) is revolutionizing modern agriculture by enabling automated monitoring and control of crops through embedded sensor networks. However, long-range communication between IoT devices remains a key challenge. LoRa (Long Range) technology has emerged as a promising solution due to its low-power, long-range (up to 10 km), and cost-effective two-way communication capabilities. While LoRa offers significant advantages, its real-world performance is influenced by environmental conditions, device parameters, and network configurations. This paper introduces FieldNet, a LoRa-powered Wireless Sensor Network (WSN) framework designed for agricultural automation. FieldNet facilitates remote, low-cost monitoring and control of distributed farming systems, addressing labor-intensive and technically demanding aspects of precision agriculture. By integrating LoRa-based WSNs, the proposed system enhances scalability, energy efficiency, and real-time data acquisition for optimized crop management. Experimental evaluations assess the impact of environmental factors on communication range, providing insights into practical deployment strategies for smart farming applications.

Index Terms—IoT, LoRa, Wireless Sensor Networks (WSN), Smart Agriculture, Precision Farming, Long-Range Communication.

I. INTRODUCTION

The advancement of agricultural technologies has become increasingly vital to meet the growing demand for food production and sustainable farming practices. Precision agriculture, which involves the use of technology to monitor and manage crops, has seen significant improvements with the integration of the Internet of Things (IoT). IoT-based solutions enable real-time data collection and automated control over various agricultural processes, thereby improving efficiency and reducing human labor. However, implementing such systems in rural or large-scale farms poses a number of challenges, particularly in terms of connectivity and communication range.

One of the major hurdles in deploying IoT in agriculture is ensuring reliable, long-range wireless communication between devices scattered across large areas. Traditional wireless communication technologies like Wi-Fi or Bluetooth have limited range and higher power requirements, making them unsuitable for remote agricultural applications. In this context, LoRa (Long Range) technology

has emerged as a promising alternative. LoRa enables low-power, long-distance communication that can reach up to 10 kilometers in open environments.

II. PHYSICAL ARCHITECTURE

In this system studies, the LoRa Network consists of three main parts: A Sensor Node, A Lora Trans-receiver, and Application.

Sensor Node-

- Collects real-time environmental data (NPK Value, soil moisture, temperature, humidity).
- Powered by batteries or solar panels for remote deployment.
- Interfaces with RISC microcontrollers for processing.

LoRa Trans Receiver-

- Responsible for long-range wireless data transmission
- Operates on Data Rate up to 300kbps & frequency bands Commonly used (433 MHz)
- Supports low-power operation to extend battery life
- Also supports sleep-mode for increased durability and more device lifespan

Application And Server-

- Stores, processes, and visualizes collected sensor data using firebase
- Provides a user interface for monitoring and control
- Implements automation alerts based on thresholds value of sensor

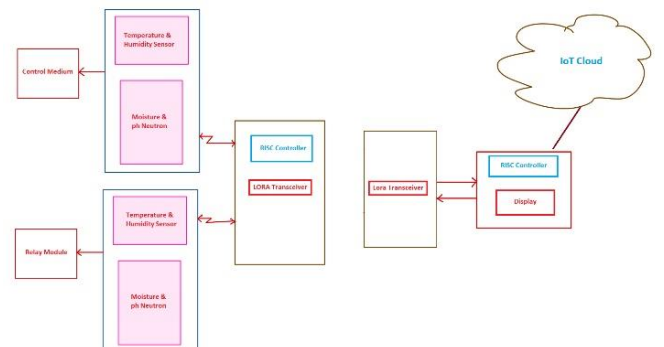


Fig 1. Block Diagram

Fig 1. shows a block diagram of the proposed system. The system consists of two sensor units. Each sensor unit includes a Temperature & Humidity Sensor and a Moisture & pH Neutron. One sensor unit is connected to a Control Medium, while the other sensor unit is connected to a Relay Module. Both sensor units are interfaced with a block that contains a RISC Controller and a LORA Transceiver. one unit transmit the sensor data and another unit received data wirelessly it consists of a Lora Transceiver. This transceiver sends the data to a final firebase server also they display data in TFT Display. The transceiver is connected to the IoT Cloud for remote monitoring and data storage.

III. COMPONENT OVERVIEW

RISC Microcontroller



Fig 2. RISC Board

The ESP32 module, low-cost, RISC-based microcontroller developed by espressif systems. Fig 2. It is widely used in IoT, embedded systems, and wireless communication projects due to its dual-core processing, Wi-Fi, Bluetooth, and low-power capabilities.

- 1 The ESP32 uses a 32-bit RISC (Reduced Instruction Set Computing) CPU (Xtensa LX6).
- 2 Dual-core variant (ESP32-D0WD) allows multitasking with two cores running at up to 240 MHz
- 3 Efficient power consumption, making it ideal for battery-operated IoT devices.
- 4 Wi-Fi (802.11 b/g/n): Supports station mode (client), access point (AP) mode, or both.
- 5 Bluetooth (BLE 4.2 + Classic): Enables short-range wireless communication.
- 6 520KB SRAM (for fast data processing).
- 7 448KB ROM (stores bootloader and core functions).

LoRa Module

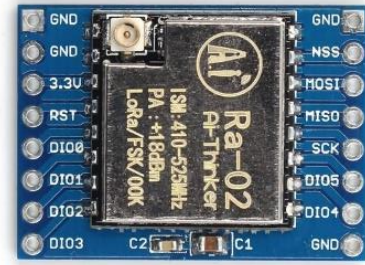


Fig 3. LoRa Module

The LoRa SX1278 RA02 works with SPI communication protocol so it can be used with any micro microcontroller that supports SPI communication protocol. It is mandatory to use an antenna along with the module else it might damage the module permanently. The module should be powered only with 3.3V, and the operating voltage is 3.3V, and the frequency is 433 MHz and transmits and receives packets up to 256 bytes.

- Frequency Range: 433MHz
- Supply voltage: 1.8~3.6V
- Standby current: 1uA
- Modulation: FSK/GFSK/MSK
- Output Power: +20dBm
- Sensitivity: -136dBm

NPK Sensor

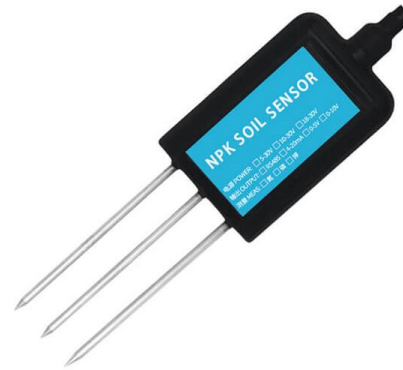


Fig 4. NPK Sensor

- What is an NPK Sensor?

Measures Nitrogen (N), Phosphorus (P), and Potassium (K) levels in soil. Essential for precision agriculture, hydroponics, and soil health monitoring. Works on electrochemical, optical, or capacitive sensing principles.
- Power supply: 3v to 5v
- Maximum power consumption: $\leq 0.15W$
- Operating temperature: $-40 \sim 80^{\circ}C$
- NPK parameters: Range: 0-1999 mg/kg(mg/L)
- Resolution: 1 mg/kg(mg/L)
- Response time: $\leq 1S$
- Protection grade: IP68

IV. NODE & RECEIVER OVERVIEW

In the first node, the ESP32 microcontroller gathers environmental data by interfacing with multiple sensors and modules.

The DS18B20 temperature sensor measures soil temperature and sends digital signals to the ESP32 through a single GPIO_14 pin, aided by a pull-up resistor to ensure stable communication. The RS485 module communicates with the Soil NPK sensor using UART protocol through the TX and RX pins of the ESP32, enabling the microcontroller to receive soil nutrient levels. The OLED display connected via I2C visually presents real-time sensor data. The LoRa Ra-02 module, which communicates with the ESP32 via SPI.

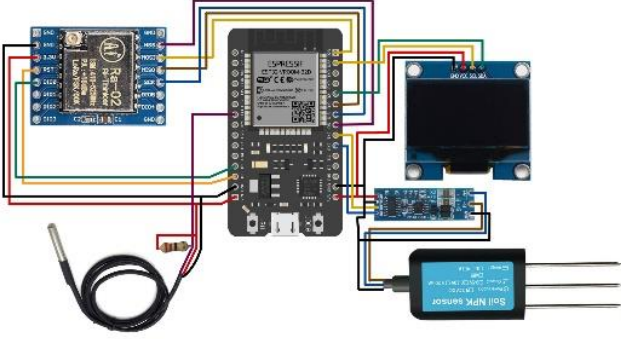


Fig 5. NODE_1

In the second node, the ESP32 also serves as the central control unit, interfacing with sensors and modules similarly to the first node.

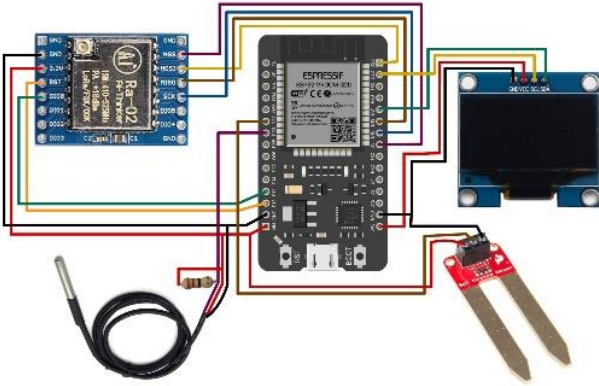


Fig 6. NODE_2

It collects temperature data from the DS18B20 sensor and soil moisture readings from the capacitive soil moisture sensor connected to an analog GPIO_32 pin. The OLED display shows the real-time values for easy local monitoring. The LoRa Ra-02 module handles wireless data transmission using the SPI interface. This node essentially monitors soil conditions like moisture and temperature.

This node functions as a centralized station for monitoring, displaying, and logging data transmitted by

multiple remote. The ESP32 receives incoming data from other nodes through the LoRa Ra-02 module, which is connected via the SPI interface. This data is processed by the ESP32 and displayed on a 1.8-inch TFT screen using another set of SPI connections, with the SD card slot potentially used for logging the incoming data. A buzzer is connected to the GPIO_14 pins and serves as an alert mechanism, possibly triggered by specific sensor thresholds.

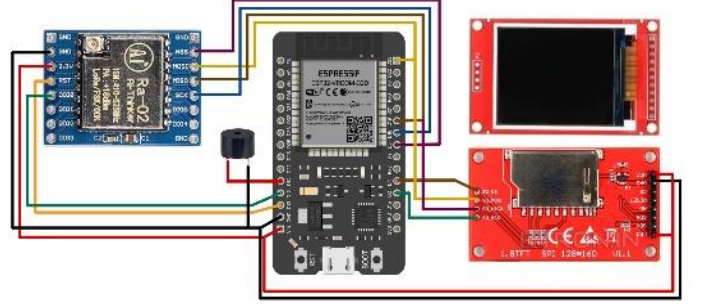


Fig 7. Transmitter-Receiver NODE

V. IMPLEMENTATION DETAILS

The entire implementation of the system consists of three nodes, each controlled by an ESP32 microcontroller in charge of data collection, transmission, and monitoring functions.

The architecture ensures efficient data acquisition along with both local and remote visualization. Local visualization is enabled through a 1.8-inch TFT display, which presents data graphically and also shows alert information. Remote visualization is facilitated via a web server. All data is stored in the Firebase cloud system. The system incorporates dual data storage, saving data locally on an SD card and remotely in the cloud, enabling comprehensive environmental monitoring with robust wireless communication. Additionally, the system provides user-friendly data presentation.

The third node receives all sensor data from remote nodes like Fig 8 over the LoRa Ra-02 module and processes the information through an ESP32 as the main transmitter-receiver station.



Fig 8. Data Collection

In Table 1 Compare the RF Module the SX1278 LoRa Module Long Transmission range up to 15 km in open areas and 3-5 km in urban environment & The Data rate supports up to 300kbps.

Module Type	Frequency Range	Data Rate
SX1278 LoRa Module	410-525 MHz (commonly 433 MHz)	Up to 300 kbps
Normal RF Module	Commonly 315 MHz, 868 MHz, 915 MHz	Up to 180 kbps
Bluetooth Module	2.4 GHz ISM band	125 kbps to 2 Mbps

Table 1. Compare RF Module

VI. RESULTS

Real-time data acquisition and remote monitoring capabilities facilitate optimized crop management, reducing labor intensity and enabling timely interventions.

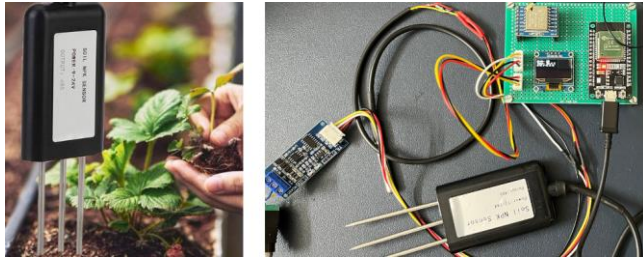


Fig 9. Sensor Data

Data is displayed graphically by converting sensor readings into visual elements such as charts. This process involves collecting data points over time, plotting them on graphs.

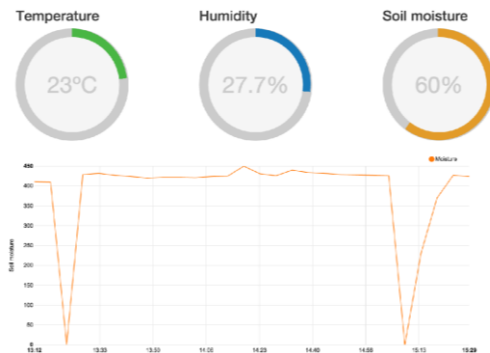


Fig10. Graphical visualization

The experimental implementation confirms that FieldNet can provide scalable, cost-effective, and energy-efficient monitoring for precision agriculture. The system's

modular architecture allows easy integration of additional sensors and expansion to cover larger farming areas.

VII. CONCLUSION

One of the primary challenges in deploying IoT solutions in agriculture is establishing reliable long-range communication across vast and often remote farm areas. Traditional wireless technologies like Wi-Fi and Bluetooth fall short due to their limited range and higher power consumption. LoRa (Long Range) technology has emerged as a promising alternative, offering low-power, long-distance communication capabilities that can extend up to 10 kilometers in open environments.

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